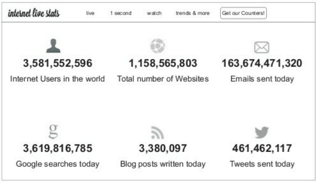

Figure 1: Real-time Big Data cases on *InternetLiveStats.com*

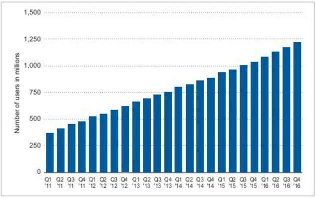

Figure 2: Number of active users per day on Facebook

helps to store and retrieve large scale files. These files can be structured or unstructured and they include documents, audio files, images, recorded video clips, binary files, etc. GridFS is similar to a file system for the storage of files. MongoDB collections are used for storage of data and files using GridFS, which has powerful features to store the files of any format, including files that are even more than 16MB in size. In classical implementations, there is the file storage limit of 16MB but MongoDB-GridFS can store and retrieve files beyond this limit too.

Using GridFS, the files are divided into a number of chunks. Every single chunk of the dataset is stored in arbitrary but logically connected documents, each with the maximum size of 255KB.

GridFS works on two collections called *fs.files* and *fs.chunks*, for the storage of metadata and a chunk of a file. Each chunk is uniquely associated with an ID. The collection *fs.files* acts as the parent document corresponding to the chunk. The field *files_id* in the collection *fs.chunks* is used to link the actual content with its parent.

The format of *fs.files* collection is:

```
{
  "filename": "********",            // Filename
  "chunkSize": "********",           // Size of Chunk
  "uploadDate": ISODate(""********"),     // Timestamp
  "md5": ""********",     // MD5 (Message Digest) Hash of File
  "length": "********",     // Size of document in bytes
   "contentType": "********",         // File Type in MIME
```

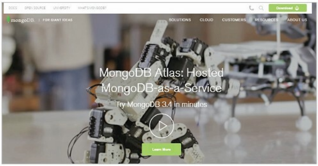

Figure 3: Portal of the MongoDB NoSQL database

```
  "metadata": "********    // Additional Information
}
```

The format of *fs.chunks* collection is:

```
{
  " _id": ObjectId(""********",""), // Unique ID of the chunk
  "files_id": ObjectId(""***",""), // Parent ID of the
document
  "n": "********",              // Sequence Number of chunk
  "data": ""********","        // Data in chunk
}
```

Adding and retrieving large binary files using GridFS

In the following example, the storage of a video file will be implemented using GridFS. The *put* command is used for this. The utility *mongofiles.exe* in the *bin* folder is required to be executed.

First, open the command prompt. Next, change the directory to the *bin* folder of MongoDB. Now start the MongoDB server by executing *mongod.exe*.

Next, execute the following command:

```
C:\MongoDBDirectory\bin\mongofiles.exe -d gridfs put MyVideo.avi
```

In the instruction above, *mongofiles.exe* is the utility for executing different commands.

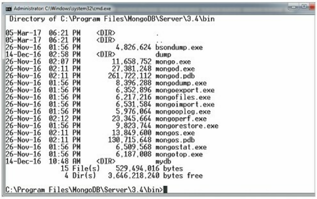

Figure 4: Utilities in the BIN directory of the MongoDB server